Chapter 23: Economic, Environmental, and Societal Issues in Materials Science and Engineering

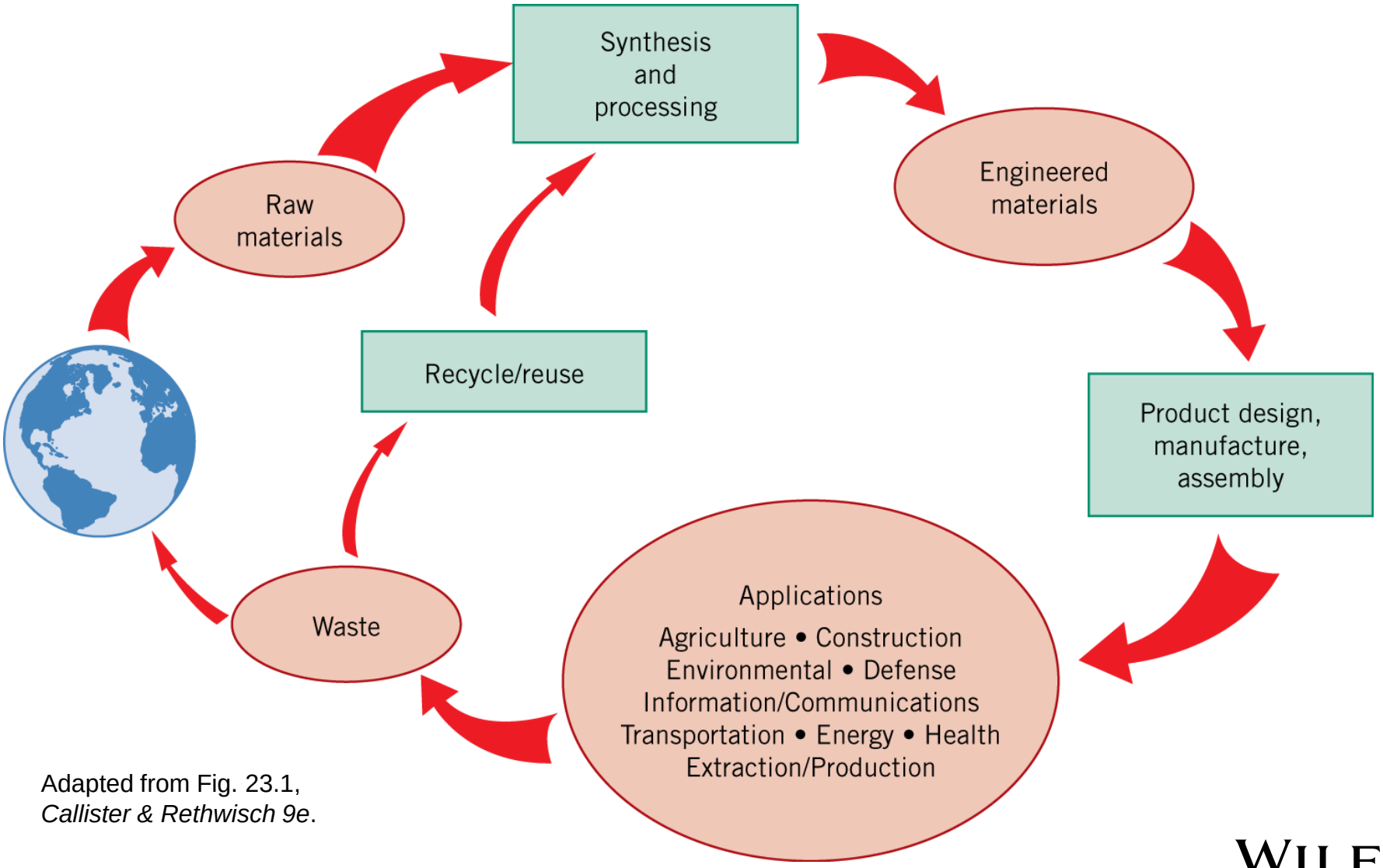
ISSUES TO ADDRESS...

- What factors affect product cost?
- What factors determine the overall environmental impact of a product?
- For which materials is recycling a viable option?
- What is “green design”?

ECONOMIC CONSIDERATIONS

- A product must make economic sense:
  -- the price must be attractive to customers
  -- it must return a sustainable profit to the company
- To minimize product cost materials engineers must consider three factors
  -- component design
  -- material selection
  -- manufacturing techniques
- Other significant factors include labor & fringe benefits, insurance, and profit

TOTAL MATERIALS CYCLE



Adapted from Fig. 23.1, *Callister & Rethwisch 9e*.

COMPONENTS OF "GREEN DESIGN"



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- **Reduce** – redesign the product to use less material
  example: PET bottles with thinner walls
- **Reuse** – fabricate the product of a material that can reused
  example: refillable bottles and shipping containers
  example: grind up old tires for use as mulch
- **Recycle** – reprocess the material into a new product
  example: convert PET bottles to carpet fibers

RECYCLING MATERIALS

- Proper product design facilitates recycling
- Advantages to recycling
  - reduced pollution emissions
  - reduced landfill deposits



Askin Durson
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- Recycling Issues
  - Product must be disassembled or shredded to recover materials
  - Collection and transportation costs are significant factors in recycling economics



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RECYCLING OF METALS

- Aluminum is the most commonly recycled metal
- Compared to refining raw ore, reprocessing metals
  - is more energy efficient
  - produces less waste (pollution)
- Difficult to recycle metals that are susceptible to Corrosion
- Toxic metals (e.g., Cd and Hg):
  - must be handled as hazardous waste
  - are difficult to reprocess
  - should not be added to landfills



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RECYCLING OF GLASS

- Glasses are the most common commercial ceramics
- Little economic incentive to recycle glass
  - raw materials inexpensive
  - relatively dense - expensive to transport
  - must be sorted by
    - color – clear, amber, green, brown
    - type – plate vs. container
    - composition – soda-lime, leaded, borosilicate



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Dale Reardon
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RECYCLING POLYMERS

- Thermoplastic polymers easily recycled
  - grind into pellets, melt, and extrude or mold into new product
  - must be sorted by polymer type – polyethylene, polystrene, PET, etc.
  - properties degrade in each reuse
- Thermosetting polymers more difficult to recycle
  - can be ground up and use as filler
  - depolymerize to make monomer
- Polymers are combustible so they can be burned to reclaim fuel value

RECYCLING COMPOSITES

- Difficult to recycle because they contain an intimate mixture of materials – difficult to separate
- Some composites are recyclable
- Components must separated by shredding or dissolution

BIODEGRADABLE POLYMERS (i)

- Polymers that degrade naturally in the environment
  - generally attacked by microbes
- Best suited for products with short lifecycles
  - Example: plastic tableware, beverage containers, golf tees
  - Example: degradable mulch films
    - retain heat and moisture
    - after harvesting crop
      - plow the film into the soil
      - decomposes into nutrients

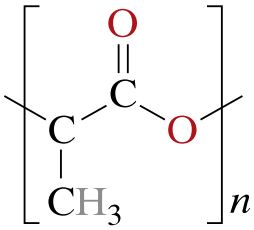


Fig. 23.3, *Callister & Rethwisch 9e*.
(Photograph courtesy of Dubois Agrinovation)

BIODEGRADABLE POLYMERS (ii)

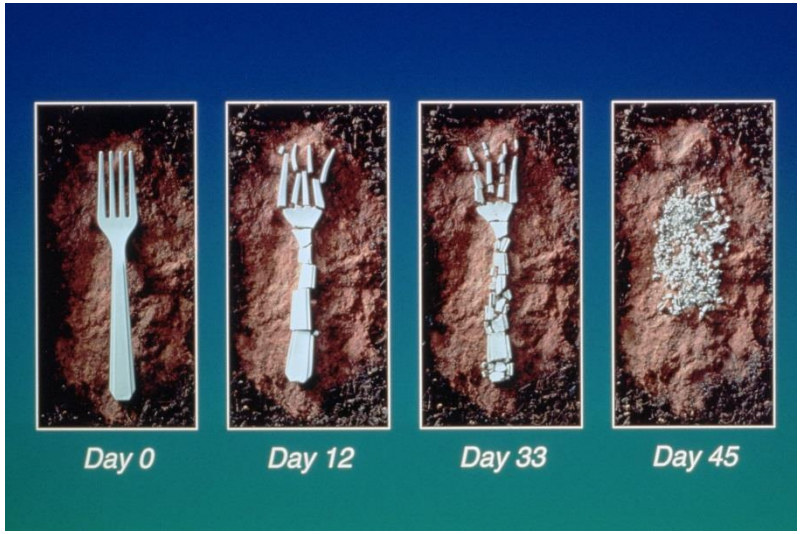
- Current generation based on biorenewable materials such as poly(lactic acid) (PLA)

- PLA structure





From p. 823, *Callister & Rethwisch 9e*. (Photograph courtesy of Natureworks LLC and International Paper, Inc.)



Chapter 23 opening photo (b), *Callister & Rethwisch 9e*. (Courtesy of Roger Ressmeyer/© Corbis)

- Long shelf life at ambient conditions
- Decomposes in months in commercial composting operations

SUMMARY

- Important for materials engineers to consider:
  -- component design
  -- materials selection
  -- manufacturing process
- Environmental and societal impacts of production are significant engineering design issues
- Cradle-to-grave life cycle assessment of products involves
  -- extraction
  -- synthesis/processing
  -- product design/manufacture
  -- application
  -- disposal
- Recyclability and disposability issues are important in materials science and engineering
- Ideally, a material should be at best recyclable, and at least biodegradable or disposable

ANNOUNCEMENTS

Reading:

Core Problems:

Self-help Problems: